

1. Java Lambda Expressions

This Java 8 challenge tests your knowledge of [Lambda expressions](#)!

Write the following methods that *return a lambda expression* performing a specified action:

1. PerformOperation isOdd(): The lambda expression must return if a number is odd or if it is even.
2. PerformOperation isPrime(): The lambda expression must return if a number is prime or if it is composite.
3. PerformOperation isPalindrome(): The lambda expression must return if a number is a palindrome or if it is not.

Input Format

Input is handled for you by the locked stub code in your editor.

Output Format

The locked stub code in your editor will print lines of output.

Sample Input

The first line contains an integer, (the number of test cases).

The subsequent lines each describe a test case in the form of space-separated integers:

The first integer specifies the condition to check for (for Odd/Even, for Prime, or for Palindrome). The second integer denotes the number to be checked.

5

1 4

2 5

3 898

1 3

2 12

Sample Output

EVEN

PRIME

PALINDROME

ODD

COMPOSITE

PROGRAM

```
import java.io.*;
```

```
import java.util.*;
```

```
public class Solution {
```

```
    interface PerformOperation {
```

```
        boolean check(int a);
```

```
    }
```

```
    public static PerformOperation isOdd() {
```

```
        return new PerformOperation() {
```

```
            public boolean check(int a) {
```

```
                return a % 2 != 0;
```

```
            }
```

```
        };
```

```
    }
```

```
    public static PerformOperation isPrime() {
```

```
        return new PerformOperation() {
```

```
public boolean check(int a) {
```

```
    if (a < 2) {
```

```
        return false;
```

```
    }
```

```
    for (int i = 2; i <= Math.sqrt(a); i++) {
```

```
        if (a % i == 0) {
```

```
            return false;
```

```
        }
```

```
    }
```

```
    return true;
```

```
}
```

```
};
```

```
}
```

```
public static PerformOperation isPalindrome() {
```

```
    return new PerformOperation() {
```

```
        public boolean check(int a) {
```

```
            int original = a;
```

```
            int reverse = 0;
```

```
            while (a > 0) {
```

```
                reverse = reverse * 10 + a % 10;
```

```
                a = a / 10;
```

```
            }
```

```
return original == reverse;
```

```
}
```

```
};
```

```
}
```

```
public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);
```

```
int T = sc.nextInt();
```

```
while (T-- > 0) {
```

```
int condition = sc.nextInt();
```

```
int number = sc.nextInt();
```

```
if (condition == 1) {
```

```
System.out.println(
```

```
isOdd().check(number) ? "ODD" : "EVEN"
```

```
);
```

```
}
```

```
else if (condition == 2) {
```

```
System.out.println(
```

```
isPrime().check(number) ? "PRIME" : "COMPOSITE"
```

```
);
```

```
}
```

```
else if (condition == 3) {
```

```
System.out.println(
```

```
isPalindrome().check(number)
```

```
? "PALINDROME"
```

```
: "NOT PALINDROME"
```

```
);
```

```
}
```

```
}
```

```
sc.close();
```

```
}
```

```
}
```

OUTPUT

The screenshot shows a web browser window with the URL <https://www.hackerrank.com/challenges/java-lambda-expressions/problem>. The page features a green banner with the text "Congratulations" and "You solved this challenge. Would you like to challenge your friends?". Below the banner, there are three test case results:

- Test case 0**: Compiler Message: Success
- Test case 1**: Input (stdin):
1 5
2 1 4
3 2 5
4 3 898
5 1 3
6 2 12
Expected Output: (empty)
- Test case 2**: Input (stdin):
1 5
2 1 4
3 2 5
4 3 898
5 1 3
6 2 12
Expected Output: (empty)

Each test case result has a "Download" link next to it. The browser's taskbar at the bottom shows the system clock as 10:06 on 12-09-2026, with a temperature of 30°C and weather "Mostly sunny".